

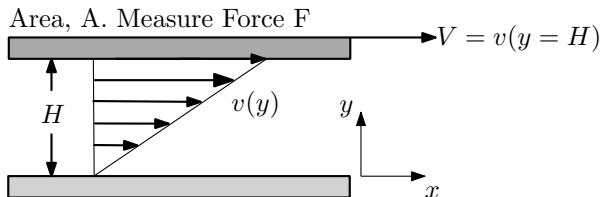
Viscoelastic Instabilities

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- ▶ Some Background on Viscoelasticity
- ▶ Some Examples of Viscoelastic Instabilities and their mechanism
- ▶ The Cross-Slot Instability

A Newtonian Fluid in Simple Shear



$$\dot{\gamma} = \frac{V}{H} \quad \text{is the shear rate}$$

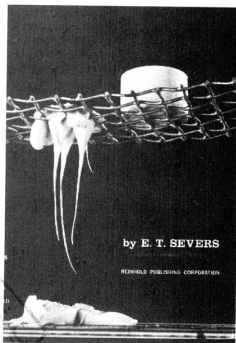
$$\tau = \frac{F}{A} \quad \text{is the shear stress}$$

$$\text{For a Newtonian Fluid } \tau = \mu \dot{\gamma}$$

Viscoelastic: More than Elastic + Viscous

- ▶ Something between a viscous liquid and elastic solid
- ▶ **but not just** elastic solid + viscous liquid.

Example: Memory Effects



Elastic Solid on "small" time scales, but viscous liquid on "large" time scales.

Small compared to what?

Dimensionless Ratios

- ▶ Deborah Number - Importance of Fluid Memory

$$De = \frac{\lambda}{T},$$

for a relaxation time (λ) and time of observation (T)

- ▶ Weissenberg Number - Importance of Elasticity

$$Wi = \frac{\text{Elastic Stresses}}{\text{Viscous Stresses}}$$

- ▶ Reynolds Number - Importance of Inertia

$$Re = \frac{\text{Inertial Forces}}{\text{Viscous Forces}}$$

Normal Stresses

In Simple Shear $\mathbf{u} = (\dot{\gamma}y, 0, 0)$, the stress tensor is:

$$\sigma = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & 0 \\ \sigma_{xy} & \sigma_{yy} & 0 \\ 0 & 0 & \sigma_{zz} \end{pmatrix},$$

with off-diagonal components, representing **tangential viscous stresses** and diagonal components, representing **normal stresses**.

In a Newtonian fluid, the normal stresses are equal to (minus) pressure: $\sigma_{xx} = \sigma_{yy} = \sigma_{zz} = -p$

In a Viscoelastic fluids, the normal stresses are not equal:

$$N_1 = \sigma_{xx} - \sigma_{yy}, \quad N_2 = \sigma_{yy} - \sigma_{zz}.$$

N_1 dominant in polymers is a **tension** in the streamlines.

The Physical Origin of N_1

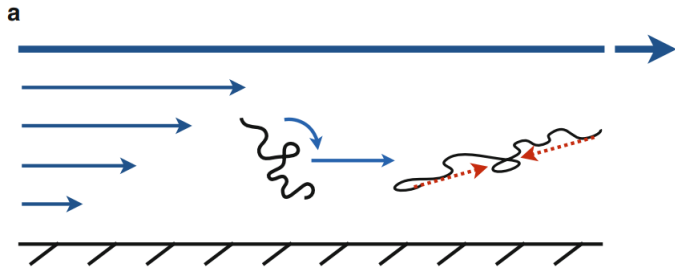
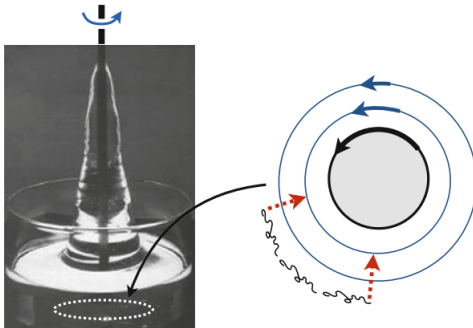


Figure: Polymers get Stretched to align with local shear

Weissenberg Effect / Rod-Climbing



Tension in Streamlines $\rightarrow$ Hoop Stress $\rightarrow$ Pushes Fluid in and up

Fluid Instabilities

- ▶ Most steady solutions are unstable to (infinitesimal) perturbations
- ▶ Idea: Start with a base state $\mathbf{U}$ and give it a kick $\mathbf{U} + \epsilon \tilde{\mathbf{u}}$
- ▶ Look for normal mode perturbations $\tilde{\mathbf{u}} = \hat{\mathbf{u}} e^{ikx + \sigma t}$ and look for growing solutions.
- ▶ Example: Taylor-Couette Instability:

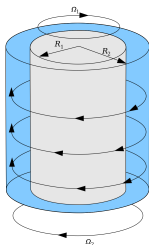


Figure: Taylor Couette Cell

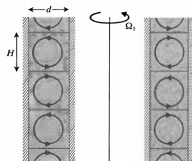
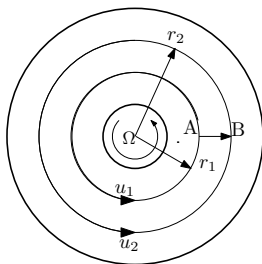


Figure: Taylor Vortices

Inertial Taylor-Couette Instability - Mechanism



In steady state:

$$-\frac{U^2}{r} = -\frac{1}{\rho} \frac{\partial p}{\partial r}$$

When moved from A to B
conservation of Angular
Momentum gives:

$$u_1 r_1 = u'_1 r_2 \Rightarrow u'_1 = r_1 u_1 / r_2$$

$u'_1 > u_2$ gives instability

Elastic Taylor Couette - Mechanism

Instability persists even in the absence of inertia ($Re=0$), due to normal stress differences.

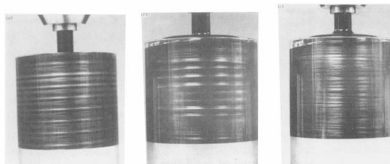
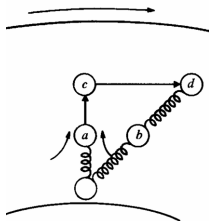


Figure: Taylor Vortices in a) a Newtonian Fluid and an elastic fluid at $Re=0$ during b) shear-startup and c) fully developed shear flow



Hoop stress when polymers get stretched along curved streamlines.

Opposing Radial flow out of phase $\rightarrow$ oscillation

Instabilities in the Plastic Industry: The Tiger Stripe Instability

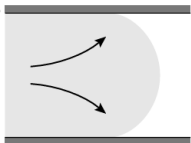


Figure: Fountain Flow



Figure: Tiger Stripe Instability

The Co-Extrusion Instability

This is caused by jumps in N_1 across an interface.

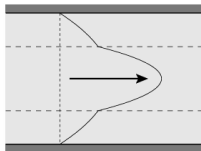


Figure: Co-Extrusion

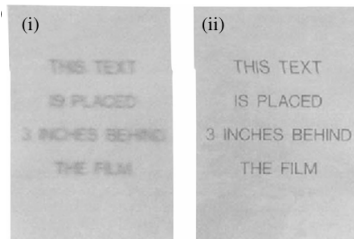


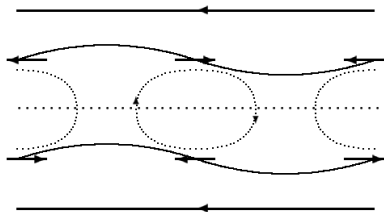
Figure: An impaired Optical Film

Mechanism behind the Co-Extrusion Instability

Suppose the outer fluid is more elastic than the inner fluid and perturb the interface.

Jump in $N_1 \rightarrow$ discontinuity in $\sigma \cdot n \rightarrow$ Secondary Flow

Mass Conservation $\rightarrow$ Recirculation



A small subtlety: a continuous variation in N_1 may not lead to instability (Wilson and Rallison 1999).

INVITED ARTICLE

Open mathematical problems regarding non-Newtonian fluids

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Figure: An Open Problem in 2012

Symmetry Breaking Instability in the Cross-slot

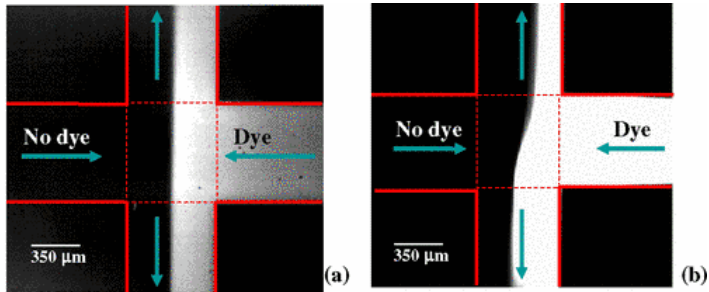


Figure: Experiments at low-Reynolds number for a) Newtonian fluid and b) polymer solution from Arratia (2006)

Immediate Questions

- ▶ Is this a real instability or an experimental anomaly?
- ▶ Can this be predicted numerically (where we can "switch off" inertia)?
- ▶ What is the mechanism behind this instability?

A Pitchfork Bifurcation

Let δ be the proportion of fluid that leaves the top outlet and originates from the left inlet.

$$DQ = \delta - (1 - \delta) = 2\delta - 1$$

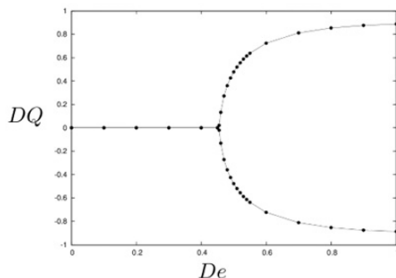
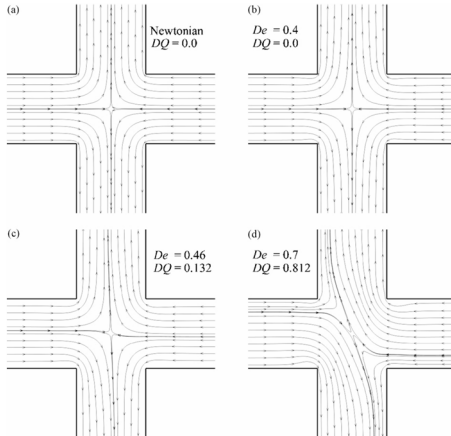


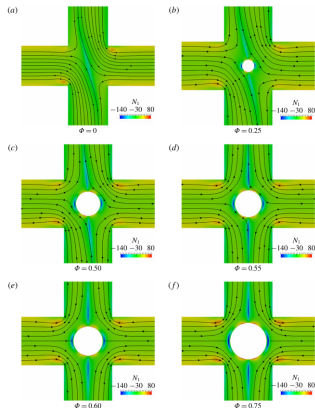
Figure: A textbook pitchfork bifurcation

Mechanism - Curved Streamlines or Stagnation Point?



- ▶ Curved Streamlines near corners
- ▶ Stagnation Point Changes nature during instability

Controlling the Instability : The Stabilising Addition of a Cylinder

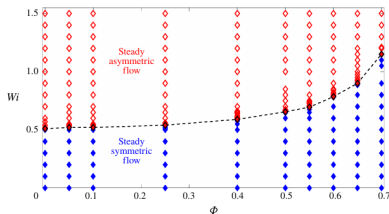


Cylinder changes the nature of the stagnation points. ($\dot{\gamma} = 0$ at stagnation points).

Figure: From Poole (2019)

Verdict?

Plot the critical Weissenberg number against blockage ratio Φ :



Notice that for small Φ , the critical Weissenberg number is unchanged, despite the loss of the high strain rates at the stagnation points.

Conclusions

- ▶ Viscoelastic fluids in complex geometries develop instabilities at low Reynolds Numbers (in the absence of inertia) due to the additional physics in their microstructure
- ▶ Understanding the mechanism of instabilities provide insight into controlling strategies

- Thank You -